Foundations of Computer Programming: Computational Thinking

Learning objectives:

- Developing a Comprehensive Understanding of Core Computational Thinking Concepts
- Understand Problem Decomposition
- Understand Pattern Recognition
- Understand Abstraction
- Understand Algorithm Thinking

Lesson exercise: Week 2

In this exercise, you will work on understanding the core computational thinking concepts and how they can be applied to problem-solving.

1. What is computational thinking?
2. The four cornerstones of computational thinking?
3. How to apply computational thinking to problem solving?
4. Computational thinking test questions:

I. Which of the following is NOT a core technique of computational thinking?

   a. Debugging
   b. Decomposition
   c. Pattern recognition
   d. Abstraction

II. The process of breaking down a complex problem into smaller, more manageable parts is called:

   a. Abstraction
   b. Decomposition
   c. Pattern recognition
   d. Algorithm design

III. Focusing on the important information and ignoring irrelevant details is an example of:

   a. Abstraction
   b. Decomposition
   c. Pattern recognition
   d. Algorithm design

IV. The step-by-step solution or set of rules to solve a problem is known as:

a. Abstraction

b. Decomposition

c. Pattern recognition

d. Algorithm design

5. Try more: https://www.bbc.co.uk/bitesize/guides/zp92mp3/test

6. For the following tasks, choose a a group of 3-5 students to work on each exercise.

- Choose a different real-world activity
- Share your findings with the class.

I. Imagine you are planning a birthday party. List all the tasks required to organise the event. Break the event planning process into smaller steps.

II. Examine the following numerical sequences. Identify any discernible patterns, then predict the next number in each sequence.

a. 2, 4, 6, 8, ...

b. 5, 10, 20, 40, ...

c. 1, 4, 9, 16, ...

III. Imagine you are giving directions to a friend to get from their house to your favourite restaurant. However, instead of describing every single landmark or street corner, simplify your directions by only including the important parts.

IV. Write a simple algorithm for making a sandwich. Each step must be clear and easy to follow, as if you're teaching someone who has never made a sandwich before.

7. Homework:

You have been asked to organise a school charity event. Use the four computational thinking concepts to solve this problem.